ITAI-1371 - Intro to Machine Learning

Individual Contribution Journal

Camila Ferreira da Silva

My contribution to Lab 06 was fixing the line breaks in the provided regression data preparation code and running it. This code loads the Titanic dataset, removes rows with missing ages, selects the features and target, and splits the data into training and testing sets. I added a verification cell that displayed 0 missing ages, 571 training samples, and 143 testing samples, confirming that the data preparation worked as expected. I also answered Knowledge Check Question 1, explaining the difference between regression and classification.

Danh Nguyen

My contribution to Lab 06 was answering all three Knowledge Check questions. For Question 1, I explained that regression predicts a continuous number, like fare, while classification predicts a category, like whether a passenger survived. For Question 2, I explained that lr_model.coef_ contains one coefficient for Age and one for Pclass, and that each shows how the predicted fare changes when that feature increases by 1 while the other stays the same. For Question 3, I explained why MSE fits regression and accuracy fits classification, and why accuracy would not work for fares, since continuous predictions almost never match exactly.

Great Inioluwa Olayiwola

In this lab, I completed and debugged the full machine learning pipeline on the Titanic dataset using Python, pandas, and scikit-learn. I resolved code syntax and import issues, handled missing data, and encoded categorical features like gender into binary format for model training. I implemented, trained, and evaluated both a LinearRegression model to predict passenger fares (evaluating performance with Mean Squared Error and Root Mean Squared Error) and a LogisticRegression model to predict passenger survival (evaluating performance with Accuracy Score). Finally, I analyzed feature weights using the models' .coef_ attributes and documented key conceptual differences between continuous regression metrics and categorical classification metrics.

Karima Sadruddin

I prepared the Titanic passenger data for a classification task by encoding the Sex column as numeric values (male = 0 and female = 1). I selected Age, Pclass, and Sex as the model's input features and Survived as the target. I split the data into training and testing sets, created a LogisticRegression model, trained it on the training data, made predictions for the test set, and used accuracy_score to measure how many predictions were correct. Before submitting, I will run the notebook to confirm that the code executes and record the accuracy value shown by the output.

Quan Thach

The coefficient for age is approximately negative zero point eleven. This means that for every additional year a passenger gets older, the model drops the predicted ticket fare by about eleven cents, assuming their passenger class remains identical. This tiny negative value

indicates that age by itself has a very minimal baseline impact on the ticket pricing once you isolate it from the financial status of the passenger class tiers. The coefficient for passenger class is approximately negative thirty-three point forty-one, indicating a massive structural impact on the model. In this dataset, passenger class is recorded as an integer where one is first class, two is second class, and three is third class. Therefore, a one-unit increase in this feature actually means moving down to a less luxurious tier. This negative coefficient means that for every step down in class tier, the predicted fare drops by about thirty-three dollars and forty-one cents, assuming the passenger age remains exactly the same.